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BTHBI338

Project Report: Bookstore Management System

Description of the Software

The Bookstore Management System is a microservice-based web application designed to facilitate the management of a bookstore. It allows users to browse a catalog of books, register, and log in to access additional features. The application consists of two main microservices: the **User Service** and the **Book Service**, along with a **MongoDB** database for persistent storage. The application is designed to be deployed on Kubernetes, ensuring scalability and resilience.

Key Features:

- **User Authentication:** Users can sign up, log in, and log out.
- **Book Catalog:** Users can view a catalog of available books, including free and paid options.
- **Admin Panel:** Admins can manage book inventory and user accounts.
- **Responsive UI:** The application is accessible via web browsers and provides a user-friendly interface.

Software Architecture Design

Architecture Overview

The architecture follows a microservices pattern, where each service is independently deployable and scalable. The services communicate over REST APIs, allowing for loose coupling and flexibility.

Components:

1. User Service:

- **Responsibilities:** Handles user registration, login, and authentication.
- **Technology:** Node.js with Express framework.
- **API Endpoints:**
 - **POST /user/signup:** Register a new user.
 - **POST /user/login:** Authenticate a user.

2. Book Service:

- **Responsibilities:** Manages the book catalog, including fetching and displaying book details.
- **Technology:** Node.js with Express framework.
- **API Endpoints:**
 - **GET /book:** Retrieve the list of books.
 - **GET /book/:id:** Retrieve details of a specific book.

3. Database:

- **Type:** MongoDB.
- **Responsibilities:** Stores user and book data persistently.
- **Deployment:** Runs as a separate microservice with persistent storage.

Architecture Principles:

- **Microservices:** Each service is independently deployable and scalable.
- **RESTful APIs:** Services communicate over HTTP using REST principles.
- **Containerization:** Each service is packaged in a Docker container for easy deployment and scaling.
- **Kubernetes:** Orchestrates the deployment, scaling, and management of the microservices.

Component Mapping:

- User Service ↔ MongoDB (User Data)
- Book Service ↔ MongoDB (Book Data)

Benefits and Challenges

Benefits:

- **Scalability:** Each microservice can be scaled independently based on demand.
- **Resilience:** Failure in one service does not affect the entire application.
- **Flexibility:** Different technologies can be used for different services.

Challenges:

- **Complexity:** Managing multiple services increases the complexity of deployment and monitoring.

- **Data Consistency:** Ensuring data consistency across services can be challenging.
- **Security:** Each service must be secured individually, increasing the attack surface.

Security Considerations:

- **Authentication:** Implement JWT (JSON Web Tokens) for secure user authentication.
- **Data Validation:** Validate all incoming data to prevent injection attacks.
- **Network Security:** Use HTTPS for secure communication between services.

Mitigation Strategies:

- Implement centralized logging and monitoring to track service health.
- Use API gateways to manage traffic and enforce security policies.
- Regularly update dependencies and perform security audits.

Configuration Management Repository

The source code for the application, including the Kubernetes deployment files, is available in the following GitHub repository: Bookstore Management System Repository.

Conclusion

The Bookstore Management System demonstrates the principles of microservices architecture, providing a scalable and resilient application for managing a bookstore. The use of REST APIs, containerization, and Kubernetes orchestration ensures that the application can handle varying loads and maintain high availability. The project highlights the importance of security and data management in a microservices environment, paving the way for future enhancements and scalability.

This report provides a comprehensive overview of the Bookstore Management System, its architecture, benefits, challenges, and security considerations. It serves as a foundation for further discussions and presentations regarding the project.

Github: <https://github.com/MAHFOOZ-16/bookstore-web.git>